

Adesh Mishra

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EDUCATION

Indian Institute of Information Technology, Bhopal

Aug 2023 – Jun 2027

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.30

- Relevant Coursework: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems, Computer Networks, Software Engineering, Database Management Systems (DBMS)

WORK EXPERIENCE

Software Development Engineer Intern

Mar 2026 – May 2026

Quantinent.ai

Remote

- Engineered 3 full-stack modules (MongoDB, Express.js, React.js, Node.js) for a financial analytics dashboard, including portfolio heatmaps and automated report generation, adopted by all internal analysts within 2 weeks of deployment.
- Architected and deployed 10+ REST APIs serving portfolio analytics across 500 financial instruments; introduced request validation middleware and centralized error handling, reducing API error rate by 35%.
- Diagnosed N+1 query pattern in MongoDB aggregation and optimized indexing; profiled React render tree and memoized 3 high-frequency components, cutting median page load time by **40%** (from ~2.5s to ~1.5s).

Competitive Programming Team Member

Nov 2024 – Sep 2025

Codame Club – IIIT Bhopal

- Delivered top 15 rank among 350+ students in competitive programming through consistent contest participation, applying advanced Data Structures, Algorithms, and optimization techniques under time constraints.
- Organized **6** competitive programming contests for **150+** participants — curated problem sets, validated test cases, and managed contest infrastructure end-to-end.
- Mentored **50+** junior students through **5** technical workshops on DSA and competitive programming strategy, improving contest participation among mentees.

PROJECTS

JobQueue | Node.js, Express.js, BullMQ, Redis, PostgreSQL, Docker

[GitHub](#)

- Built a distributed job processing system with **BullMQ** and **Redis**, exposing HTTP endpoints to enqueue, monitor, and manage asynchronous jobs while persisting job metadata in **PostgreSQL**.
- Implemented **5** fault-tolerant background workers with automatic retries (up to **3** attempts), exponential backoff, and real-time job status tracking (Pending, Processing, Completed, Failed), reducing failed-job recovery time by 40%.
- Containerized the application using **Docker**, isolating API server, Redis, PostgreSQL, and worker services to enable reproducible deployments and horizontal worker scaling.

DocQuery AI | Streamlit, langchain, FAISS, Mistral-AI, PyPdf, Python

[Live](#) | [GitHub](#)

- Developed an AI-powered book Q&A assistant using Streamlit, LangChain, FAISS, Mistral-AI, supporting PDF uploads up to 100 MB and enabling context-aware answers with 90% relevance across test queries.
- Implemented an end-to-end **RAG pipeline** with PDF parsing, recursive text chunking, embeddings, and **FAISS vector search**, indexing **1000+ text chunks** for efficient semantic retrieval and grounded response generation.
- Integrated LangChain's retrieval chain, memory buffer, and prompt template modules with Streamlit session state to maintain multi-turn context, improving response relevance across follow-up queries.

Forever | React.js, Node.js, Express.js, MongoDB, Cloudinary

[Live](#) | [GitHub](#)

- Developed a full-stack clothing e-commerce platform using **React.js**, **Node.js**, **Express.js**, **MongoDB**, and **Cloudinary**, building **15+ REST APIs** to support customer and admin workflows.
- Built **JWT-authenticated** customer and admin portals with **20+ reusable React components**, enabling product, inventory, cart, and order management.
- Configured Cloudinary for optimized image uploads and delivery across **200+** product listings, reducing backend storage load by 70%.

ACHIEVEMENTS

Competitive Programming: **1692** rating on CodeChef; **Specialist (peak 1421)** on Codeforces; **1200+ DSA problems** solved across LeetCode, CodeChef, and Codeforces.

Flipkart Grid: **Semi-finalist** in both **6.0** and **8.0** editions (Top 2.3%).

HackWithInfy: Qualified national-level round; selected for Specialist Programmer (SP) L3 interview.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C++, C, Python, Java, HTML, CSS

Databases: PostgreSQL, MongoDB, MySQL, Redis

Technologies: React.js, Next.js, Node.js, Express.js, Redux, Tailwind CSS, pgvector, NextAuth.js

Tools & Infra: Git, GitHub, Docker, AWS EC2, Postman, CI/CD

Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Distributed Systems, System Design, Microservices Architecture, Concurrency & Multithreading, RESTful API Design, SDLC, Version Control.